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Bearing Predictive Maintenance Model Analysis and Technical Evaluation Report

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Presentation Summary

It gives an overview of a hybrid artificial intelligence system design for the estimation of the Remaining Useful Life for industrial bearings. The main aim of the project is ensuring the shift in maintenance strategy from the reactance approach to the knowledge-based preventive approach by eliminating the associated costs and risks of Occupational Health and Safety during unplanned downtimes. The system designed also solves the main challenges in the literature approach such as Context Blindness and Instability in predictions. The most distinct innovations of the model are as follows:

- **Hybrid Architecture:** It embeds a dynamic branch-LSTM to learn from time series data regarding degradation trends, while a static branch-Dense adds context to the predictions by processing operational conditions-speed and load.
- **Sensor Fusion:** Unlike traditional approaches focusing only on vibration data, it increases physical consistency by incorporating temperature data, which is a direct indicator of mechanical wear, into the model.
- **Engineering-Oriented Stabilization:** The raw AI outputs are filtered through a two-layer filtering-the Median Filter and the EWMA-to make them stable, readable predictions that can be industrially used.
- **Safety-Oriented Approach:** the system uses as fail-safe logic the 20g acceleration limit, which is the physical safety threshold, and it develops an early warning by being on the safe side in situations where uncertainty exists.

It is evident from the results that the developed model performs with high trend alignment performance and acceptable error margins on the showcase bearings. In particular, the fact that the system is capable of successfully simulating the degradation process and providing early warnings for safety in unexpected scenarios is a strong proof of the applicability of the developed system.

System Summary

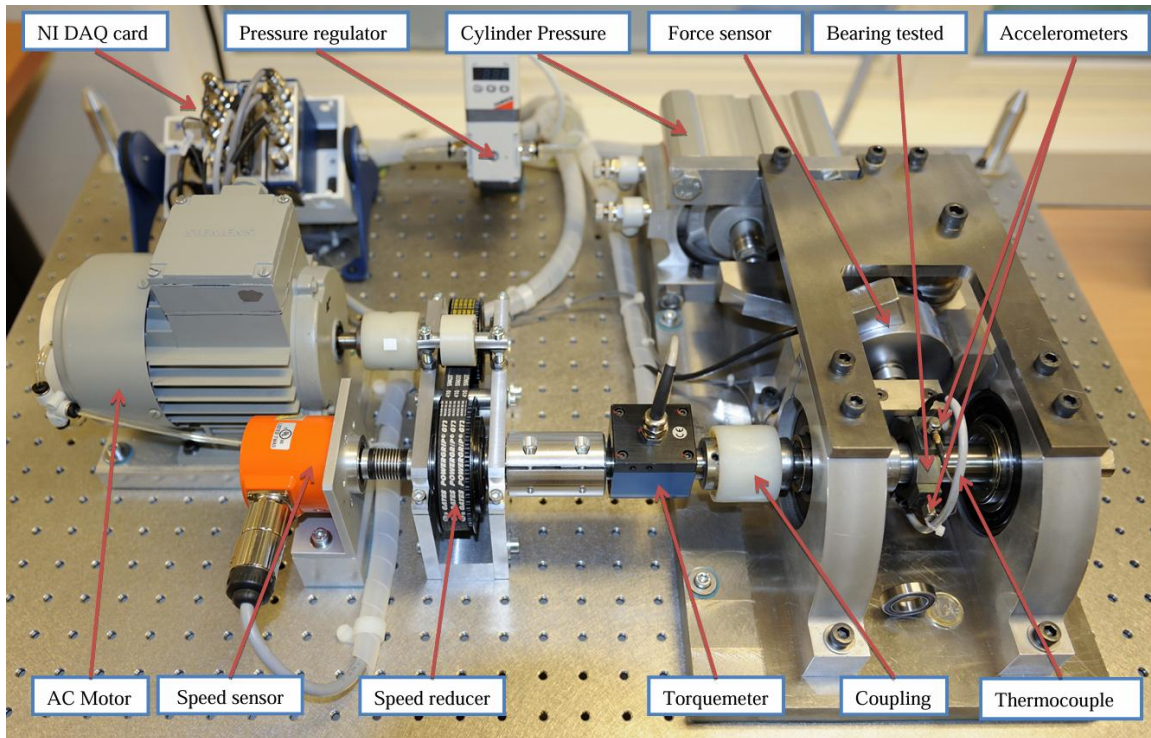


Figure 1 Overview of PRONOSTIA

To achieve these ends, a PRONOSTIA experimental environment was developed. It was created by FEMTO-ST Institute, whose main purpose is to accelerate the degradation of ball bearings. This schedule follows a run to failure methodology. This yields accurate information necessary for training a predictive maintenance algorithm. Such a test environment enables monitoring of bearings from normal operation to complete failure in a short period of time. This makes it relevant for improving bearings.

Starting with the physical aspect, the motor that propels the system has a power of 250W, which is asynchronous. This motor drives the rotation, which is then transferred using a gearbox and a coupling assembly to the testing bearing. To efficiently promote mechanical wear, a customized bearing loading system is implemented. This system incorporates a pneumatic jack with a lever arm that presses a radial force of up to 4000N directly on the outer ring of the bearing. Such high pressure significantly shortens the life span of the bearing, allowing real defects such as those of the inner or outer raceways or balls.



Figure 2 Measurement part: accelerometers and temperature sensor

Data acquisition is achieved through the utilization of the multi-sensor technique, where two distinct physical signals are targeted. Vibration signals are acquired by the incorporation of two miniature accelerometers placed at a 90-degree angle on both horizontal and vertical planes. The devices possess a very high sampling rate of 25.6 kHz to allow the detection of higher-frequency vibration characteristics. Specifically, the process involves the utilization of a periodic recording approach, where instead of streaming, there is a 0.1-second recording (2560 samples) every 10 seconds to optimize data processing. On the other hand, the recording of temperature variation in the bearing is acquired by the utilization of the PT100 RTD temperature sensor, which is mounted close to the outer ring, sampled at 0.1 Hz. All experiments involving data acquisition in the current dataset are under the termination process to prevent catastrophic failure of the test rig; data acquisition is halted when the vibration amplitude exceeds the safety threshold of 20g.

1. Industrial Problem Definition

1.1. From Reactive Care to Predictive Care

The driving factor of the project is that there are high costs associated with the reactive, post-failure maintenance practices that are common in the industry. Though unplanned downtimes lead to lost production, failures lead to a hazardous working environment. This should be addressed by developing a system that uses data-driven Predictive Maintenance techniques, which should be able to foresee any failures that are yet to occur.

1.2. Technical Objectives and Gaps in the Literature

Though the contemporary research endeavors in the field of academics result in effective outcomes when placed in laboratory settings, they are inadequate when considering the realities associated with the dynamics of the field. The proposed project will help address two major gaps that exist in the literature:

- **Context Blindness:** Conventional approaches cannot differentiate under what operational circumstances a vibration signal is created. This project tackles this challenge by merging static operational information with time-series information.
- **Lack of Physical Evidence:** Most models fail to consider temperature, which is among the most basic physical effects of mechanical wear. Temperature information is considered within this model using a sensor fusion concept.

2. Data Methodology and Feature Engineering

2.1. Data Set: IEEE PHM 2012 (PRONOSTIA)

In the project, data obtained from the PRONOSTIA experimental platform, accepted as an industry standard, were used.

- **Sensors:** In each experimental setup, two accelerometers in the horizontal and vertical axes and a temperature sensor are found.
- **Sampling Rate:** Vibration data were collected at a high frequency such as 25.6 kHz, while temperature data were collected once every 10 seconds.
- **Physical Stopping Condition:** Experiments are automatically stopped when the acceleration amplitude reaches the 20g value in order not to cause permanent damage to the setup. This 20g limit is defined as the RUL = 0 (absolute failure) point for the model.

2.2. Engineering-Oriented Data Selection

Not all of the 17 bearings in the dataset are suitable for prognostic analysis. Some bearings exhibit unpredictable behaviors such as Sudden Fracture. Therefore, since the goal of the project is to learn the evolution of wear rather than chaos, only bearings demonstrating a Monotonic Degradation Trend were selected via an engineering decision. The lineup selected for analysis and presentation is as follows:

- Bearing1_3, Bearing1_4, Bearing1_5, Bearing1_6, Bearing1_7, Bearing3_3

2.3. Feature Engineering

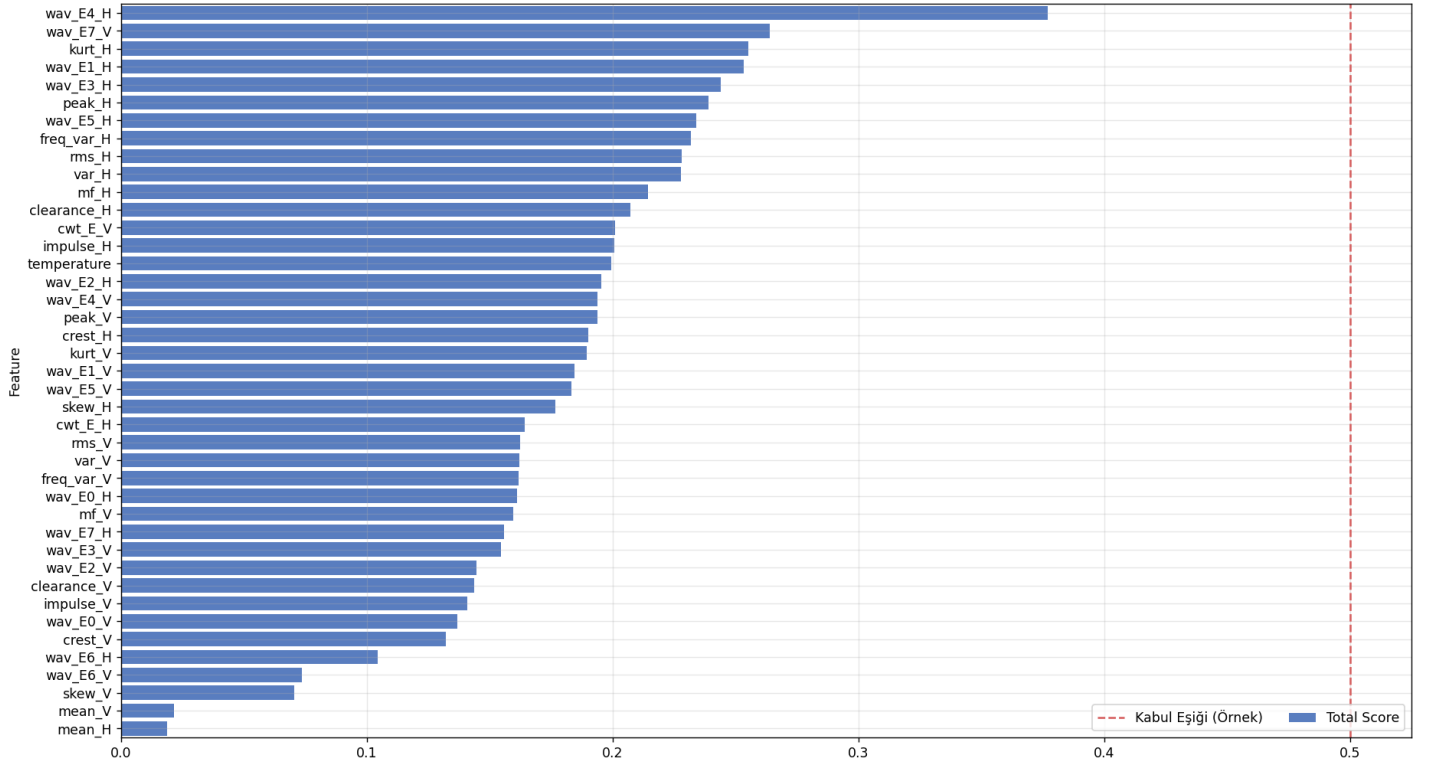


Figure 3 Parameter Value

Raw sensor data have been transformed into meaningful Health Indicators for the artificial intelligence to learn.

$$X_{rms} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$$

$$Kurtosis = \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^4}}{\sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2}^2}$$

Time Domain Analysis:

- RMS (Root Mean Square): Represents the total energy power of the signal and increases with wear.
- Kurtosis: Captures sudden impacts and micro-cracks in the signal.
- Crest Factor & Impulse Factor: Analyzes the characteristics of the peaks in the signal.

Frequency Domain Analysis (Wavelet Transform):

- Since the Standard Fourier Transform (FFT) loses time information, the Wavelet Transform, which provides both frequency and time localization, was used. In this way, it is detected in which frequency band and when the degradation begins.

Feature List: As a result of comprehensive analyses, the features determined to be the most valuable for RUL estimation were selected. This SELECTED_FEATURES list contains wavelet energies (wav_...), time domain statistics (kurt_H, rms_H, etc.), and most importantly, temperature and condition variables for physical context.

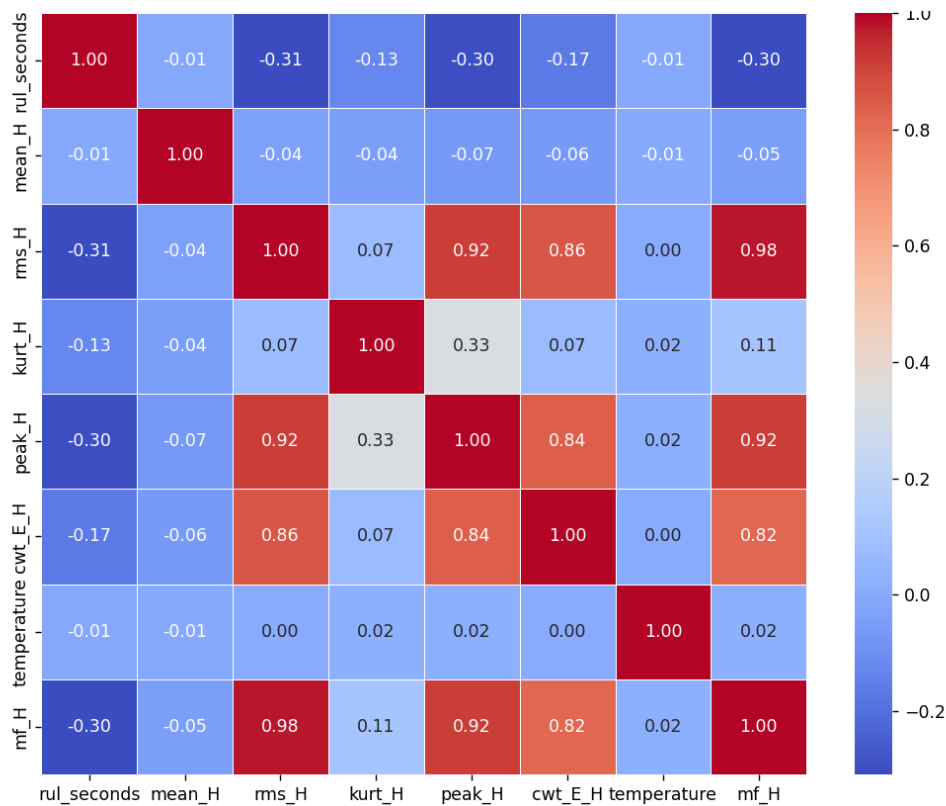


Figure 4 Attribute Feature - RUL Correlation Matrix (High negative correlation with RUL is good)

As shown from Figures 2 and 1, feature selection is not random. Although the correlation matrix shows that RMS and Peak values are the most strongly related to degradation, from variance analysis, ranking the most valuable parameters is based on their monotonicity score. The analysis removed unnecessary data to provide the model with pure information.

2.4. Data Packaging

- The data for the training of the model have been prepared in the following manner:
 - Feature Smoothing: An EWMA (Exponential Weighted Moving Average) filter ($\alpha=0.3$) was applied to the features to clean sudden spikes in the sensor data.
 - 3D Windows (Windowing): The data were packaged into 30-unit (`SEQUENCE_LENGTH = 30`) time windows so that the LSTM architecture could learn past trends.
 - Normalization: All dynamic features were scaled between 0-1 using MinMaxScaler.

3. Hybrid AI Architecture

The core of the model is a dual-arm hybrid structure that processes data from two different channels to provide context.

3.1. Components of Architecture

Dynamic Branch (LSTM Layers):

- Task: It learns the degradation trend and the "memory" of the wear by processing time-series dependent vibration and temperature features.
- Structure: It consists of two consecutive LSTM layers with 128 and 64 neurons. Stability is increased by applying BatchNormalization and Dropout (30%) after each layer.

Static Branch (Dense Layers):

- Task: It processes operational conditions (`CONDITION_PARAMS`) such as motor speed and applied load.
- Logic: It eliminates "context blindness" by enabling the model to answer questions such as "Did this vibration occur under high load?".
- Structure: It contains a Dense layer consisting of 32 neurons and Dropout (20%).

Fusion and Decision Layers:

- Trend information coming from the dynamic branch and operational context information coming from the static branch are combined in the Concatenate layer.
- The combined data is passed through two Dense layers with 128 and 64 neurons, and the final RUL prediction is generated.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t \tanh(C_t)$$

3.2. Robust Training Strategy

Loss Function: It is quite robust to sensor noise and outliers, which are inherent in the industrial datasets. It acts like MSE when the error is small and like MAE when the error is large to keep the model as less affected by noise as possible.

Regularization: Dropout (at rates from 20% to 40%) and L2 Regularization (0.001) techniques were used to prevent the model from memorizing the training data.

Optimization:

- Optimizer: Adam optimizer (learning_rate=0.001).

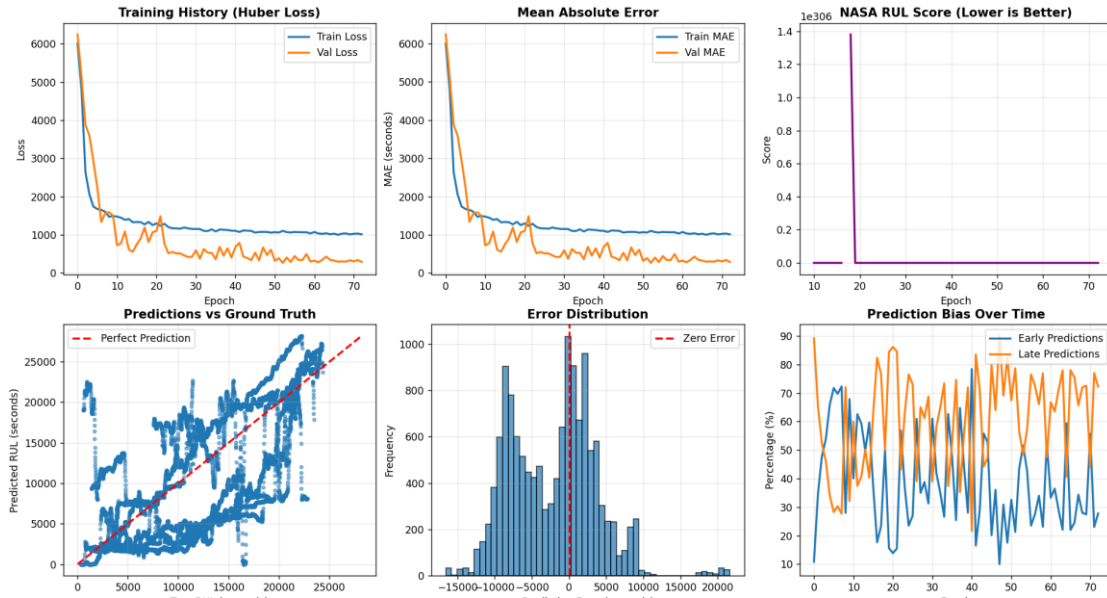


Figure 5 Mixed Results for All Bearings (Analysis of Target Bearings Below)

The declining trend in the Loss graph at the top left indicates that it is reducing the error, while indicating that the Predictions are being made in a stable manner without bending towards either side in the Prediction Bias graph. All these graphs are indications that it is not overfitting.

4. Analysis of Results and Engineering Evaluation

The model's raw estimates have undergone further processing for industrial reliability and usability..

4.1. Dual Layer Prediction Stabilization

Raw artificial intelligence outputs may contain momentary spikes that could shake operator confidence. Therefore, a two-stage "smoothing" process has been applied to the predictions:

- 1st Layer (Median Filter): It shaves off sudden and illogical peaks (spikes) resulting from sensor errors by using a 15-unit window (window=15).
- 2nd Layer (EWMA Filter): By applying an exponential weighted moving average with alpha=0.05 to the cleaned data, the trend is smoothed and a stable RUL line is obtained.,

4.2. Numerical Performance Metrics

The overall avrage RMSE from the analysis performed on the Showcase Bearings is 4858.02 seconds.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^N (y_i - \bar{y}_i)^2}$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \bar{y}_i|$$

Bearing	MAE (Hata)	RMSE	Trend Harmony (Korelasyon)
Bearing1_3	2950.19 s	3924.71 s	78.60%
Bearing1_4	3785.07 s	5632.93 s	80.00%
Bearing1_5	3030.11 s	3690.01 s	93.40%
Bearing1_6	5744.18 s	6922.80 s	77.70%
Bearing1_7	5222.79 s	5720.72 s	97.20%
Bearing3_3	2325.59 s	2919.49 s	28.30%

Table 1 Conclussions

4.3. Visual Case Studies and the "Safe Side" Approach

Ideal Trend Tracking: The model has closely followed the actual degradation curve with a very high correlation.

Pessimistic and Safe Prediction: In some cases, the model has deliberately predicted a shorter life than the actual life. This is not an error, but a safety measure providing early warning.

Challenging Conditions: Even in situations where there are sudden changes, the general degradation trend has been captured.

4.4 Signal Characteristic

To fully grasp the effectiveness of the artificial intelligence model, it is important to comprehend the nature of the raw signal the model processes. The Kurtosis levels presented by the yellow lines in Figures 4 and 5 indicate a chaotic form with sudden bursts of activity. The objective of the model is to convert the chaotic yellow wave into a smoother blue line. The effectiveness derived is the capacity to distill the signal within the chaotic environment.

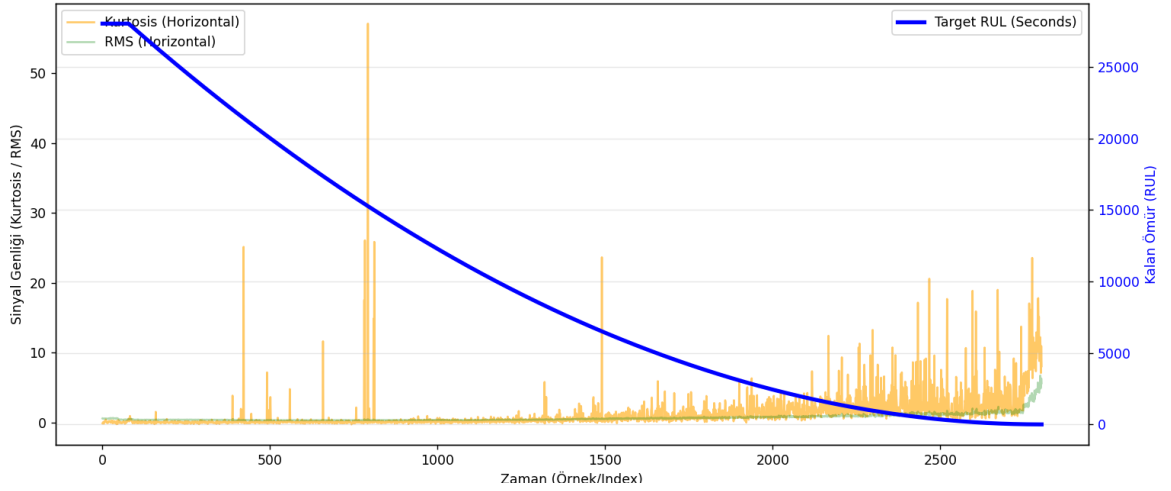


Figure 6 Bearing 1_1 Physical Signal vs. Artificial RUL Tag

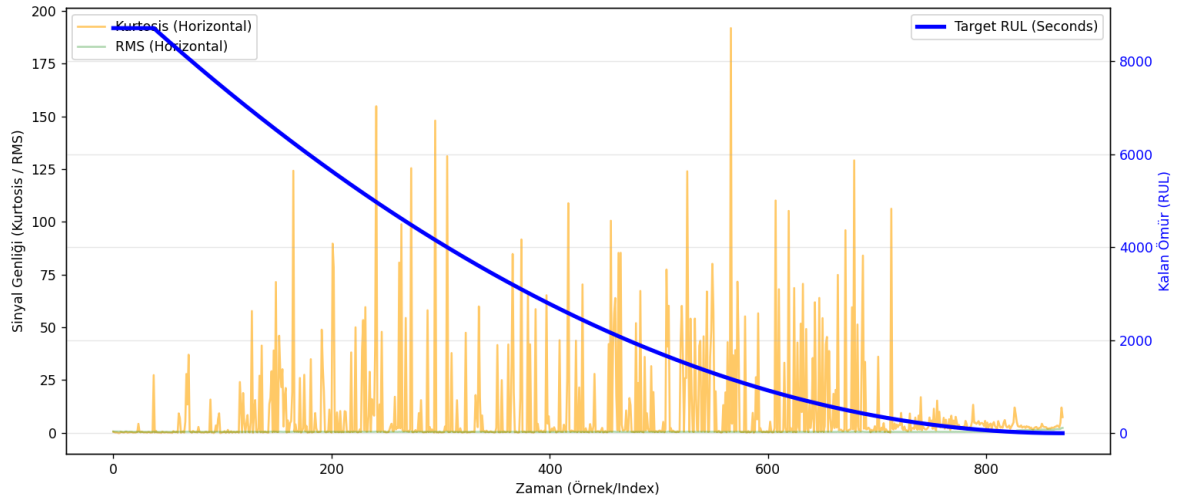


Figure 7 Bearing1_2 Physical Signal vs. Artificial RUL Tag

5. Conclusion and Future Vision

The created hybrid model is able to handle context blindness in an effective way using the static data branch and overcome prediction instability using the dual-layer stabilization technique. By using the temperature data and adding constraints like the acceleration constraint of 20g, the created model is used as an engineering assistant tool and is not a black box model because it is dependent on the data. Moreover, the model has low inference time and therefore is useful in the real-time predictive maintenance software running in industries.

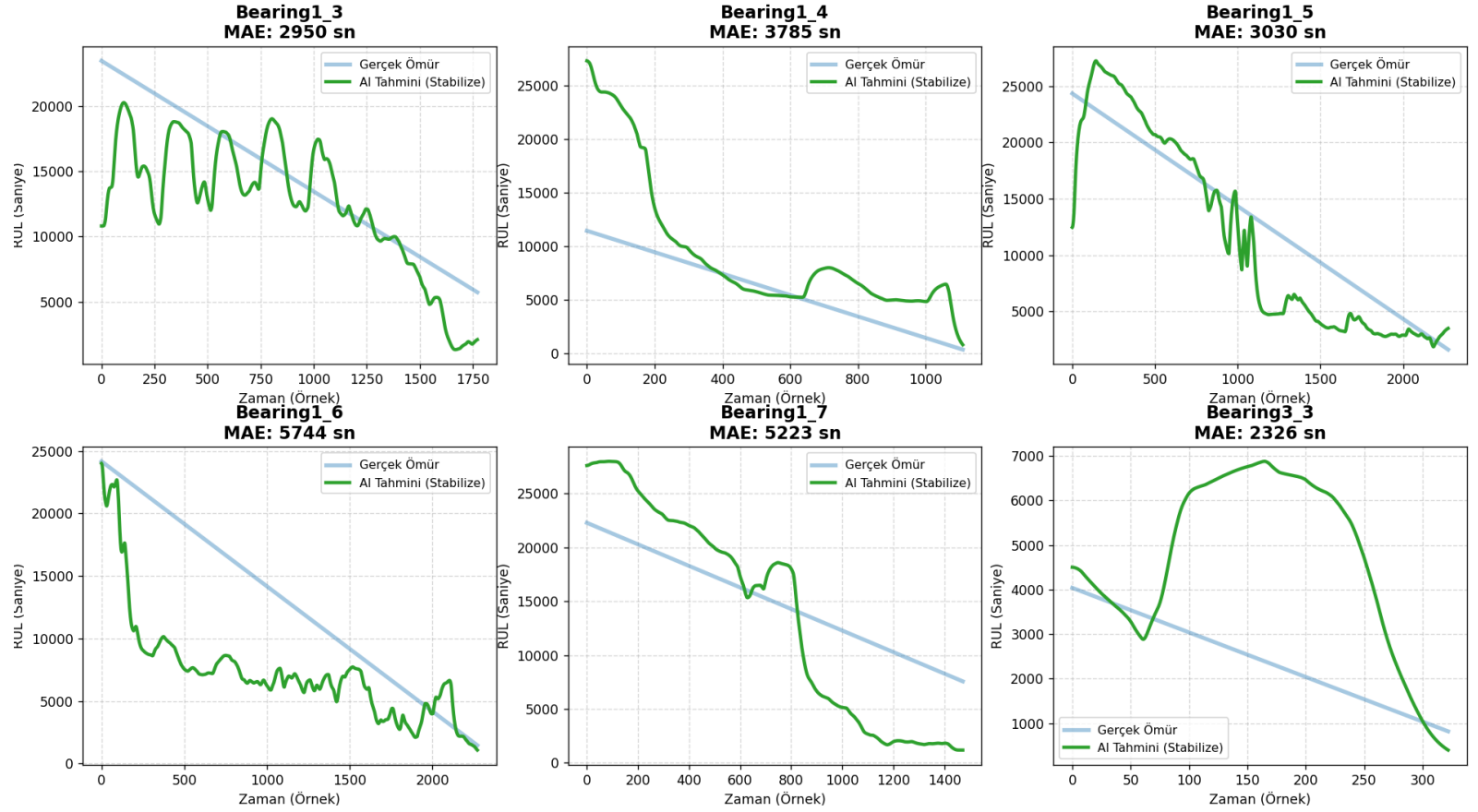


Figure 8 Instantaneous Bearing Prediction

While much of the existing literature assumes a uniform degradation process, attempting to model both sudden fracture events and long term wear mechanisms within a single architecture often compromises prediction stability. In contrast, this study prioritizes industrial reliability by optimizing the model specifically for monotonic degradation behavior. Consequently, bearings prone to sudden fracture (e.g., *Bearing 3_3*) are intentionally excluded from the current scope, as such chaotic failure modes require a distinct, class-based modeling framework. From an engineering perspective, developing specialized expert models tailored to specific degradation regimes rather than enforcing a universal model represents a more robust and scalable strategy for high-precision RUL estimation.

Model Approach	Degradation Assumption	Context Awareness	Prediction Stability	Industrial Readiness
Single-Branch LSTM	Uniform degradation	Low	Medium	Limited
Statistical Model	Strictly monotonic	None	High	Low
Universal (Sudden + Wear)	Mixed regimes	Medium	Low	Risky
Proposed Hybrid Model	Monotonic wear	High	High	High
Sudden Fracture (Future)	Sudden fracture	Medium	Low-Medium	Case-dependent

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